TNE20002/TNE70003 - Network Routing Principles

Portfolio Task – Scenario 5
Pass Task

Introduction

This Network Routing Principles **Scenarios** are a scaffolded approach to preparing you to succeed in your ultimate **Final Skills Assessments**. The **Scenarios** build on skills from previous **Scenarios** until all required components are covered. **Scenario 5** expands your work to cover deployment of **DHCP** on the Internal Network and **NAT** on the gateway router of the Internal Network. For **Scenario 5-P**, you will essentially repeat your work from **Scenario 4-P** to consolidate your knowledge in deployment of Interior Routing Protocols and ACLs before expanding on this in the **C/D** Tasks and in **Scenario 6-P**.

Purpose

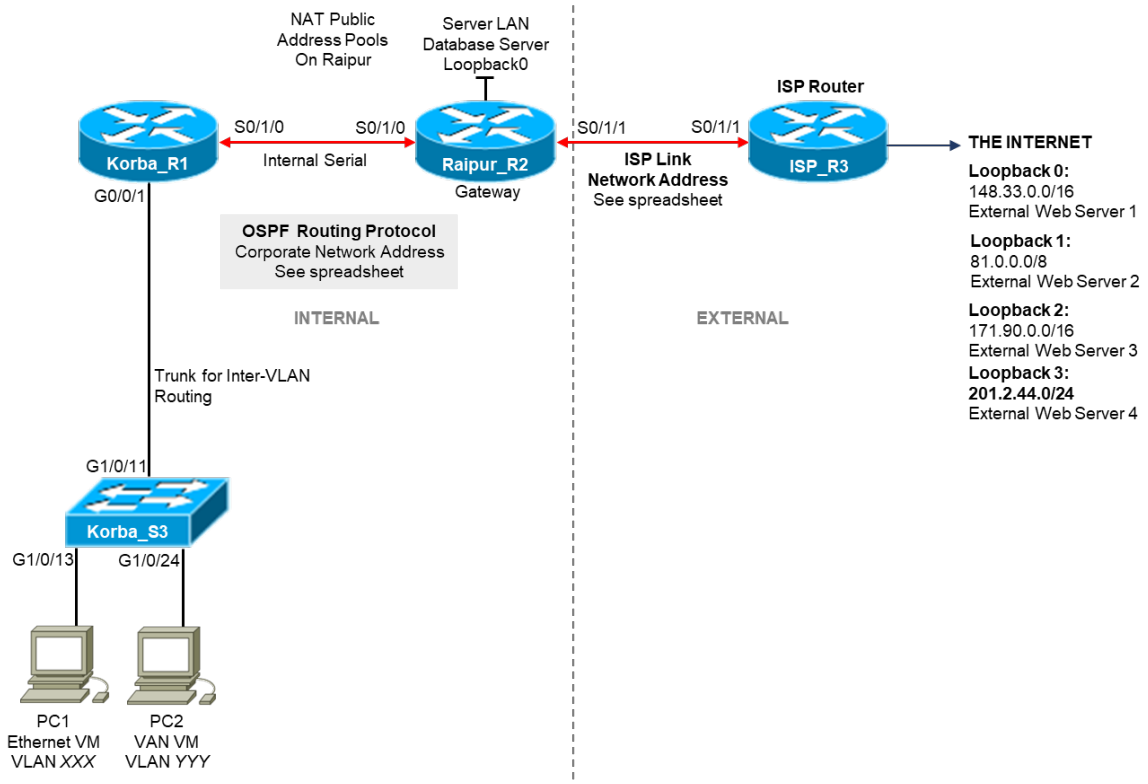
In this **Scenario** you will design and construct a network consisting of three routers and one switch, matching the hardware configuration of your Final Skills Assessment. You will consolidate the skills you acquired in building an internal network using a Routing Protocol connected to an external network via a public IP address coupled with ACLs to protect segments of your network. In this **Scenario** you will be repeating existing work in constructing a base network to later introduce new skills. **No new** tasks will be covered in **Scenario 5-P**.

Methodology

This portion of the handout contains the necessary information to design and build your network. Information on the assessment is at the end of the handout.

Network Topology

The Network topology is displayed in the figure below.



Network Information

The Network topology diagram refers to a number of network addresses and VLAN names. Please use the **provided spreadsheet on Canvas** to obtain your personalized network information for **Scenario 4**. The spreadsheet will provide:

- Corporate Network Address
- ISP Link Network Address
- **VLANXXX**, **VLANYYY**, and **VLANZZZ** VLAN Identification

Subnetting

The first task you must perform is to subnet your Corporate network to create subnets for your VLANs. The subnetting requirements are:

Network	VLAN Name/Interface	Connected Switches	Host Count
VLANXXX	Green	Korba	1,500 hosts
VLANYYY	Blue	Korba	20 hosts
VLANZZZ	Grey	Korba	180 hosts
VLAN1	-	Korba	14 hosts

Internal Serial Links	-	-	2 hosts
Database Server LAN	Loopback 0	Raipur	30 hosts

Please have a copy of your working in case it is needed during assessment. You will need to document your assignment of IP addresses to Router Interfaces and PC Hosts

NOTE: You may use a subnetting Calculator to calculate the subnets but you should be able to do it more quickly without one

Basic Network Configuration

You are essentially rebuilding the network from **Scenarios 4** , however there are fewer networking devices and subnets for you to construct. You will still be configuring the network using the **OSPF Routing Protocol**. Please refer to the previous Scenario Instructions, or more specifically your Lab Journal, if you need assistance in meeting the following requirements.

- Check physical wiring on the devices
- Configure a MOTD and Hostnames on all devices
- Set the MOTD banned to include your student ID, name, and Lab time
- Configure the Switch with an enable password of **cisco**, the necessary VLANs, a management interface on VLAN1, a default gateway, and telnet access with password **cisco**
- Configure Switch ports G1/0/13 and G1/0/14 as access ports on VLANXXX with port security settings of (mac address sticky, max 4, violation protect), and port G1/0/24 as an access port on VLANYYY
- Configure all serial and loopback addresses on routers with interface descriptions
- Configure all routers connected to the switch with inter-VLAN routing using a trunk connection to the switch
- On the ISP router, configure only a static route to the Internal network

Before continuing, you should run all necessary tests to confirm that all the requirements listed above are properly configured.

OSPF Requirements for Scenario

For the purposes of the Scenario, you must configure OSPF on the internal routers as per the instructions below:

- Run OSPF on all internal corporate routers
- Configure the bandwidth for the point-to-point links between routers as:
    - **Raipur**-**Korba**– configure bandwidth 512
- Advertise all internal network addresses on all internal routers, advertising each subnet individually with an appropriate wildcard mask
- Advertise the default route installed on the gateway router – **Raipur**

- Disable broadcasting on internal edge-networks (all interfaces connected PCs) – all sub-interfaces of **g0/0/1** on **Korba**

ACL Requirements for Scenario

The ACL security requirements for this Scenario are:

Generic ACLs

1. PCs in VLAN XXX **permitted** HTTP access to ISP Loopback 0 and deny ALL other access to this interface.
2. PCs in VLAN XXX **denied** PING requests to PCs in VLAN YYY
3. PCs in VLAN XXX **permitted** PING replies to PCs in VLAN YYY
4. PCs in VLAN XXX **permitted** ALL access to the Internet.
5. ALL access to the Internet – all the other Servers.

NOTE: Requirements 2 and 3 above mean that PCs in VLAN YYY are able to ping PCs in VLAN XXX BUT that PCs in VLAN XXX CANNOT ping PCs in VLAN YYY.

Telnet ACLs

1. **ONLY** PCs in VLAN XXX **permitted** TELNET access to **Korba** Router
2. **ONLY** PCs in VLAN XXX **denied** TELNET access to **Raipur** Router

Assessment

The Scenario is assessed in class by your Lab Supervisor. When you have successfully configured and tested the Scenario, you will need to demonstrate functionality to your Supervisor. Upon successful demonstration, the Supervisor will ask you 1 or 2 questions about the Scenario in order to confirm that you completed the work and not another student. Upon successfully answering these questions, the Scenario will be marked as complete.

The due date for Scenario 5 is at the start of the Lab in Week 10. As a pass task, you MUST successfully complete this task in order to be eligible to pass the Portfolio and therefore the Unit.

NOTE: The final date for assessment of Scenario 5 is in Week 10. Failure to complete by Week 10 will result in failing this task and potentially failing the Unit.

What Happens if I Fail

Failure in this task will result in you **failing** the Unit. Students who reach the end of semester with **ONLY one** missing/failed **Pass-Level** Scenario will be afforded an opportunity to re-submit and enable you to Pass the Portfolio. **Students with two or more incomplete Pass Level Scenarios will fail the portfolio and therefore the Unit.**